

Coalescence-Induced Jumping for Removing the Deposited Heterogeneous Droplets: A Molecular Dynamics Simulation Study

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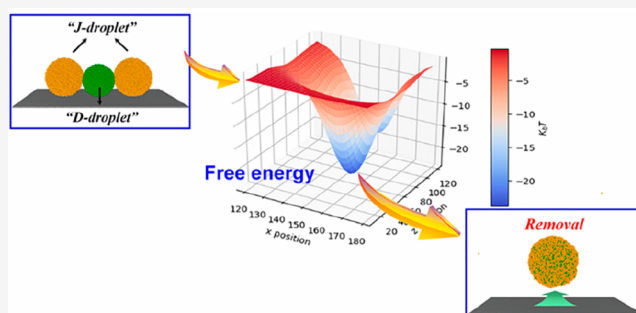


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Supporting Information

ABSTRACT: The removal of the deposited droplets on a solid surface is crucial to considerable practical applications that require self-cleaning properties. In this work, a strategy of cleaning a deposited droplet (“D-droplet”) by coalescing with a heterogeneous and easily jumping droplet (“J-droplet”) is proposed. Molecular dynamics simulation studies have shown that the coalescence of these two kinds of droplets would not guarantee the removal of D-droplet, unless the lifting ability of J-droplet is enhanced through the reduction of the solid–liquid interaction. However, this is a bad scenario with low efficiency. Further investigation suggests that by introducing two J-droplets to produce triple-coalescence dynamics, the D-droplet could be successfully jumping from the substrates due to the coalescence-induced effect, which is also verified by the free energy calculation. Moreover, the effects of the size of the droplets and the arrangement mode of these three droplets on the jumping dynamics are both considered. The studies not only help advance our understanding of coalescence-induced jumping of heterogeneous droplets, but also open up new ways to remove the deposited impure droplets, which is expected to guide the fields of self-cleaning.



INTRODUCTION

The coalescence dynamics of droplets are common phenomena encountered in our daily life. Once two droplets contact, they will instantly merge into one bigger droplet under the driving force of reduction of the surface energy, accompanied by the release of the excess surface energy. When this process occurs on superhydrophobic substrates or low-adhesion substrates, the coalesced droplet can jump away from its surface as the obtained kinetic energy transformed from the excess surface energy overcomes the liquid–solid interaction.^{1,2} Coalescence-induced droplet jumping has profound implications for various biological and industrial applications, such as self-cleaning,^{1,3–5} antifogging,⁶ anti-icing,^{7–9} water harvesting,¹⁰ enhanced condensation heat transfer,^{11,12} and corrosion protection,^{13,14} which has attracted more and more research attention.

It is known that the hydrodynamics of the liquids on a solid substrate is easily and greatly affected by the surface properties,¹⁵ especially the surface structures,^{16–19} so a large amount of research focusing on the modification of surface structures has been done to better control the coalescence-induced droplet jumping behaviors from both the theoretical simulation and experimental investigation. For example, by carrying out molecular dynamics (MD) simulations, Gao et al.²⁰ investigated the coalescence process of two equal sizes of droplets on microtextured surfaces decorated with pillared structures, and found that the decrease of surface wettability intensity and solid fraction could reduce the attraction force between the droplets and substrates, and thereby easily induce the coalescence-

induced jumping behavior. As in another simulation work, He et al.²¹ explored the effect of curved surfaces on the coalescence dynamics. Compared with the flat smooth surface, the surface curvature could enhance symmetry breaking and improve energy conversion efficiency, which leads to a higher jumping velocity of the coalesced droplets. Recently, an asymmetric V-groove was designed on the superhydrophobic surfaces for achieving more than 700% increase in energy conversion efficiency, which not only effectively enhances the velocity of coalescence-induced droplet jumping, but also randomly changes the jumping direction by controlling the groove angle.²² Additionally, some experimental studies also reported many well-designed structures that are useful to promote the self-propelled jumping behaviors of the coalesced droplet, such as the nanogras zigzag structures,²³ hierarchical microcone structures,^{24,25} and V-shaped grooves.^{26,27}

In addition to the surface structures, the droplet itself also plays a critical role in the coalescence-induced jumping behaviors. One concern first raised is the size effect. In earlier times, it has been found that only for the droplets with a certain

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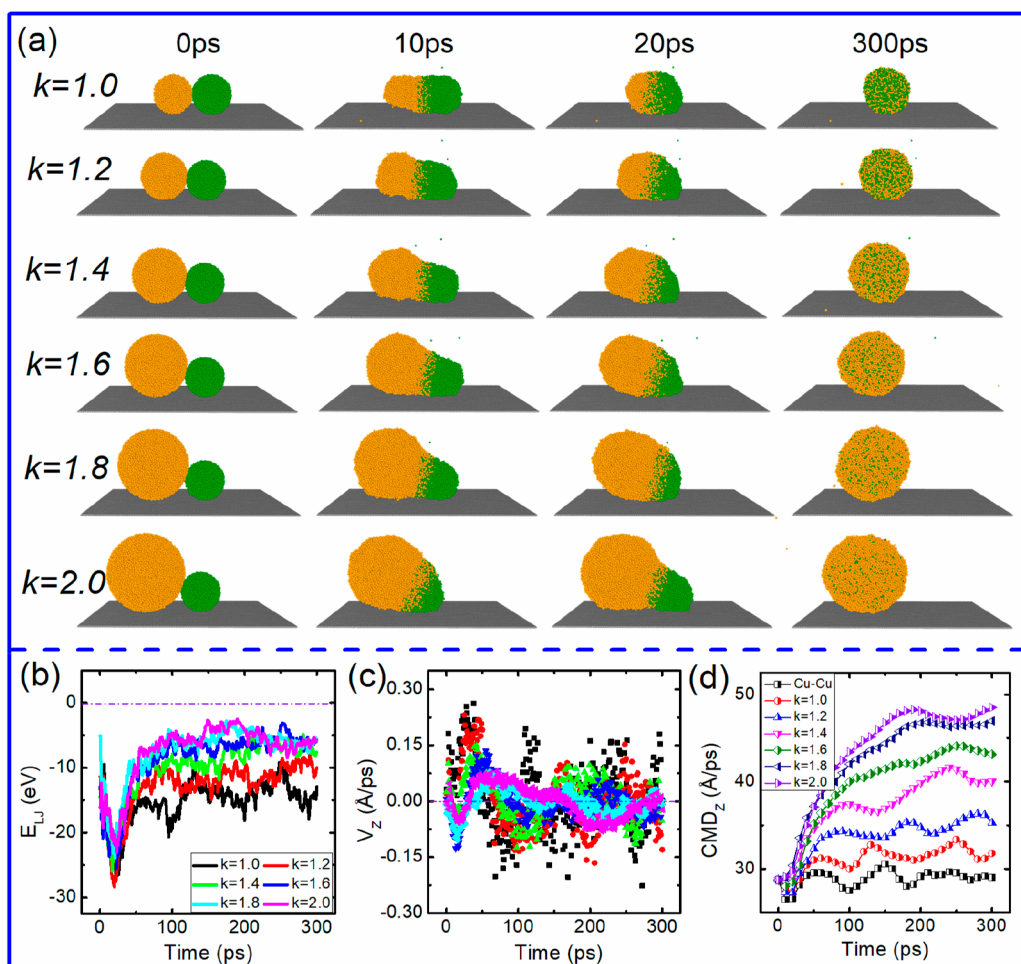


Figure 1. (a) Heterogeneous coalescence behaviors of Cu droplets ($R_2 = 25$ Å) and Ag droplets with different size, k is defined as R_1/R_2 , $k = 1.0, 1.2, 1.4, 1.6, 1.8, 2.0$. (b) Interaction energy between the liquids and the substrate with time. (c) Movement velocity of the coalesced droplets along the z direction (V_z). (d) Center-of-mass displacement of the Cu droplets along the z direction (CMD_z) during the coalescing process.

range of radius can the coalesced-jumping behaviors occur.²⁸ This study still considers the droplets with the same size, and later coalescence dynamics of two unequal-sized droplets which are commonly encountered in reality are investigated. Wang et al.'s work demonstrated that when the size ratio between a smaller droplet and a bigger droplet is larger than 0.56, no jumping would be transformed into a jumping regime.²⁹ Wasserfall et al.³⁰ have further studied the size ratio of the droplets with a wide range of wettability for determining regimes of jumping and adhesion, and pointed out that the further reduced energy conversion efficiency for unequal-sized droplets caused by strong asymmetric flow behavior would be the reason for weakening the jumping behaviors. Further studies have been carried out in a theoretical and numerical way, and a generalized inertial-capillary scaling law was established for describing the coalescing behaviors of droplets with equal or unequal size.³¹

Although great progress was made in controlling or understanding the coalescence-induced jumping behaviors, existing studies mainly focused on the homogeneous droplets, while coalescence of heterogeneous droplets has gotten less attention. The asymmetric movements of two different droplets must affect their coalescing dynamics,³² which is difficult to predict by the current laws or models established for two identical droplets. For different kinds of liquid droplets, their adhesion with a certain solid surface is largely different. It is

learned that coalescence-induced jumping is favored when liquid–solid interaction is smaller, but fails for the high-adhesion case. So what would happen if a jumping droplet coalesces with a deposition one, and could we use an easy-jumping droplet to remove the deposited impure droplet via coalescing behaviors? Given that, in this work, we would like to investigate the coalescence-induced jumping of two different droplets with various adhesion properties. The aim of this research is to achieve the goal of lifting the previously deposited droplet by coalescing with the easy-jumping one, which may provide theoretical implications for some applications, especially in the field of self-cleaning.

MODELS AND METHODS

Molecular dynamics (MD) simulations have been adopted to study the coalescence-induced jumping behaviors of the heterogeneous droplets (a basic model can be found in Figure S1), by using a large-scale atomic/molecular massively parallel simulator (LAMMPS) package. The reason why we use metallic droplets for the study is shown in the Supporting Information. In our simulations, NVT ensemble and periodic boundary conditions³³ are adopted. The temperature is controlled by a Nose-Hoover thermostat,^{34,35} a time step of 1.0 fs is adopted for all the simulations, and the Velocity-Verlet algorithm is used to

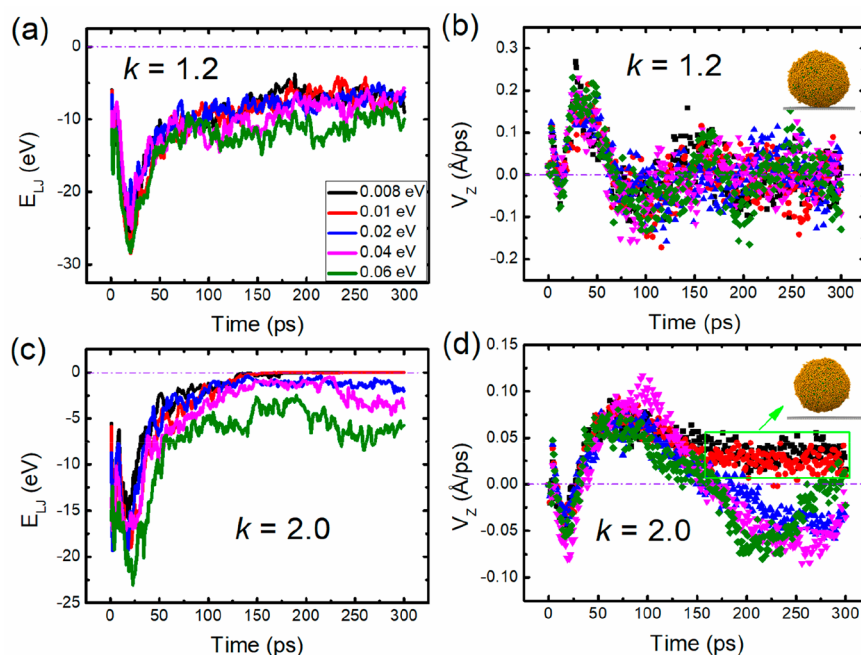


Figure 2. (a) Interaction energy between the liquids and the substrate and (b) movement velocity of the coalesced droplets along the z direction (V_z) with time under different ϵ values in the case of $k = 1.2$. (c) Interaction energy between the liquids and the substrate and (d) movement velocity of the coalesced droplets along the z direction (V_z) with time under different ϵ values in the case of $k = 2.0$.

solve Newtonian motion equations.³⁶ All the simulations run about 500 ps.

Carbon atoms of the substrates are described by the adaptive intermolecular reactive empirical bond order (AIREBO) potential,^{37–39} which is expressed as

$$E_{\text{AIREBO}} = \frac{1}{2} \sum_i \sum_{j \neq i} \left[E_{ij}^{\text{REBO}} + E_{ij}^{\text{LJ}} + \sum_{k \neq i, j} \sum_{l \neq i, j, k} E_{kijl}^{\text{TORSION}} \right] \quad (1)$$

The interactions between metallic atoms in the droplets, i.e. Cu–Cu interaction, Ag–Ag interaction, and Cu–Ag interaction, are modeled by embedded atom method (EAM) potential,^{40,41} which can be written as

$$E_{\text{EAM}} = F_{\alpha} \left(\sum_{j \neq i} \rho_{\beta}(r_{ij}) \right) + \frac{1}{2} \sum_{j \neq i} \phi_{\alpha\beta}(r_{ij}) \quad (2)$$

where F is the embedding energy and ϕ is a pair potential interaction. Finally, the interactions between liquid atoms and substrate atoms are described by 12–6 Lennard-Jones (L-J) potential, given as

$$E_{\text{L-J}} = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right], \quad r < r_c \quad (3)$$

where ϵ is the depth of the potential wall, σ is the finite distance at which the interparticle potential is zero, r_c is the cutoff distance of 10.0 Å. The related parameters are $\epsilon_{\text{Ag-C}} = 0.006$ eV, $\sigma_{\text{Ag-C}} = 2.5$ Å;⁴² $\epsilon_{\text{Cu-C}} = 0.01$ eV, and $\sigma_{\text{Cu-C}} = 3.225$ Å.⁴³

Free energy calculations were implemented in SSAGES software.⁴⁴ The advanced sampling is based on the Metadynamics simulation.⁴⁵ Metadynamics defines a class of flat-histogram methods useful for MD simulations. In metadynamics, a history-dependent bias is accrued through the periodic

application of elemental Gaussian biases to the collective variables. The form of the Gaussians biases is

$$g(\xi, s_i) = W_i e^{-(\xi - s_i)^2 / 2\sigma_i^2} \quad (4)$$

where s is the position of the bias, ξ is the collective variable, W is the height, and σ is the width of the bias. As the simulation proceeds, when a sufficient number of biases have been applied in a given region of the CV space, the total applied biases will begin to oscillate around the negative of the free energy. Therefore, the free energy surface could be calculated by summing over all applied biases:

$$A(\xi, t) = -V(\xi) = - \sum_{i < n(t)} W_i e^{-(\xi - s_i)^2 / (2\sigma_i^2)} \quad (5)$$

where $n(t)$ is the total applied biases. In this work, two collective variables (CVs) were used, x and z position of the center of mass for a group of atoms in the Cu droplets. The variation of free energy in different positions could help us to understand and explain the droplet dynamics.

RESULTS AND DISCUSSION

The deposition of the droplets in many fields is unexpected, such as self-cleaning anti-icing,^{7,46,47} and antifogging,^{48,49} and great effort has been made to remove the droplet by homogeneous coalescence-induced jumping behaviors. In the above studies, however, it is seen that this strategy fails to remove the “D-droplet”, as proven by Figure S2. Therefore, we here propose another strategy, that is, using the “J-droplet” to lift the D-droplet by coalescence. Working from this idea, the coalescence dynamics of Cu droplets and Ag droplets are studied, as shown in Figure 1. The Cu droplets are considered as the D-droplet with a fixed size of $R_2 = 25$ Å, while the Ag droplets are used as the J-droplet with the size R_1 ranging from 25 to 50 Å. A parameter k is defined as the ratio of R_1 to R_2 , namely, $k = R_1/R_2$, to quantitatively compare the relative size of these two droplets.

From Figure 1(a), it can be found that the final coalesced droplets are deposited on the surface in all cases, suggesting the failure of removing the D-droplet in the current research. And the velocity of the coalesced droplets along the z direction is around zero, without forming a stably positive value for supporting the jumping behaviors, as indicated in Figure 1(c). Figure 1(b) shows the time-dependent liquid–solid interaction energy, which remains at the nonzero values, further indicating that the droplets stay on the substrate. However, it is noted that with the size increase of the Ag droplets, the interaction energy between the coalesced droplet and substrate decreases, which denotes a larger detachment tendency and a stronger lifting ability. This is testified by the CMD_z of the Cu droplet in Figure 1(d). For coalescing with a larger Ag droplet, the Cu atoms in the coalesced droplet would be lifted to a higher place. The fact indicates that with further increasing the value of k , the “D-droplet” may be removed from the surface. But this idea would be difficult to be achieved when the size of the given droplets is only in a certain range, so we would like to seek another strategy for solving this issue.

It is known that the decrease of the liquid–solid interaction could enhance the efficiency of coalescence-induced jumping as more kinetic energy would be obtained from the reduction of the surface energy of two droplets during coalescence. So decreasing the interaction between J-droplets and substrate is adopted to enlarge their lifting ability for the purpose of removing D-droplets. As mentioned in the Model and Methods section, the Ag–C interaction energy is described by the 12–6 L-J potential, which is determined by two parameters, ϵ and σ . By changing the value of ϵ , the Ag–C interaction energy could be directly tuned for research. The suitable parameter of ϵ for accurately describing the Ag–C interaction is 0.06 eV, which has been testified by our previous study.⁴² We here set a series of $\epsilon_{\text{Ag-C}}$ values (0.04, 0.02, 0.01, 0.008 eV) smaller than 0.06 eV for increasing the detachment ability of J-droplets and study their coalescence behaviors with D-droplets. Figure 2 shows the variation of E_{LJ} and V_z with time under different values of ϵ . In the case of $k = 1.2$, the liquid–solid energy for all values of ϵ never becomes zero (Figure 2(a), indicating the coalesced droplets are still touching the substrates. Figure 2(b) also shows that there is no stable positive detachment velocity for lifting the coalesced droplets. When the size of the D-droplets increases, in the case of $k = 2.0$, we find that only for the $\epsilon = 0.01$ and 0.008 eV do the energy values become zero at the latter time (Figure 2(c)), indicating that the coalesced droplets could jump from the substrates. The velocity curves in Figure 2(d) also illustrate that there is a period of time showing stable positive values of V_z for the $\epsilon = 0.01$ and 0.008 eV, which can support the detachment of the coalesced droplets along the upward direction of the substrates.

For other cases, the final results are summarized as a coalescence characteristic map shown in Figure 3. It could be observed that the J-droplets can eventually lift the D-droplets when the ϵ is further reduced, but also occurs with the larger k . Generally, it is foreseeable that if we continue decreasing the value of ϵ , the value of k for the jumping case also decreases. Although the D-droplets can be lifted based on this scenario, it is not an easy and inexpensive way for practical applications as one must design a special surface structure or introduce coating to reduce the solid–liquid interaction.

The strategy of adding another J-droplet for triple-coalescence is considered to lift the D-droplet. As shown in Figure 4, another Ag droplet is placed on the right side of the previous two

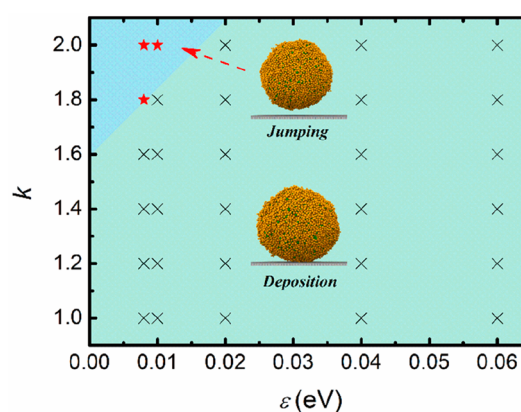


Figure 3. Coalescence characteristic map of two heterogeneous droplets with different parameters of k and ϵ .

droplets, and their center-of-mass is in one plane at the beginning. Except for the case of $k = 1.0$, it can be seen that in other cases, the D-droplets are finally lifted by coalescing with two J-droplets, suggesting that the removal of impurity liquids could be achieved in this scenario. Compared with a two-droplet system, more excess surface energy would be released after triple-coalescing, which could provide more kinetic energy for the coalesced droplets under the same conditions, so two J-droplets would be more useful to remove the D-droplet through the coalescence-induced jumping dynamics. Figure 5(a) displays the droplet–solid interaction energy with time, which further demonstrates that only the case of $k = 1.0$ fails to lift the D-droplet. For $k = 1.0$, we can also note that the droplet would stably stay on the substrate and gradually evolves into the equilibrium state as the interaction energy has little change and the V_z oscillates near zero (Figure 5(b)). Generally, once the coalesced droplet detaches from the surface, the V_z can be regarded as the jumping velocity, which is nearly unchanged with time. But the value of V_z seems to be independent of the size of the J-droplet. So the movement distance from the substrate (i.e., jumping height) would be a nonmonotonic increase with the size increase of the J-droplet, denoted by Figure 5(c). However, the change of jumping time is monotonic. As shown in Figure 5(d), it can be found that a long time would be required for the detachment of the D-droplet when the J-droplet is bigger. This is because the larger size of the J-droplet needs more time to accomplish the coalescence process before jumping, which causes a longer jumping time.

Figure 5 panels e and f present the free energy landscape of two cases ($k = 1.0$ and $k = 1.2$). We here mainly focus on the z positions for better comparing the jumping and deposition behaviors, so the z range is set from 10 to 120 Å. For the case of $k = 1.0$, we can find that there are many minimum energy values at about 30–60 Å along the z direction, indicating that these positions should be the most stable states for the Cu droplet. Obviously, this distance is unable to support the detachment of the Cu droplet, and that is why the Cu droplets could not be removed in the case of $k = 1.0$. When the k is increased to 1.2, there is a clear tendency that the free energy decreases with the z position, and the energy minima would only occur at a larger z value (larger than 60 Å), suggesting that detachment from the substrate is the most possible state of the Cu droplets. In short, free energy gives us a better view of understanding the jumping and deposition behaviors of the droplets in different scenarios.

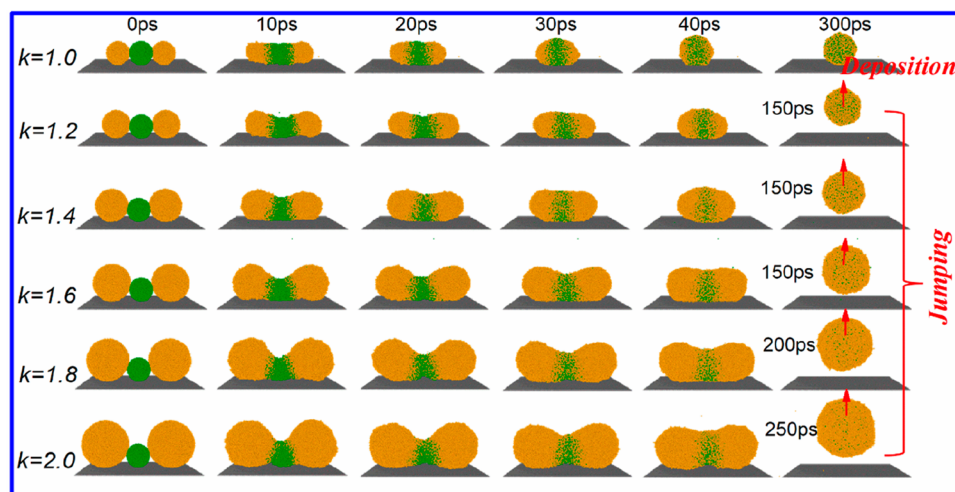


Figure 4. Triple coalescing behaviors of the Cu droplets and two Ag droplets with different sizes, while the Cu droplets are in the middle.

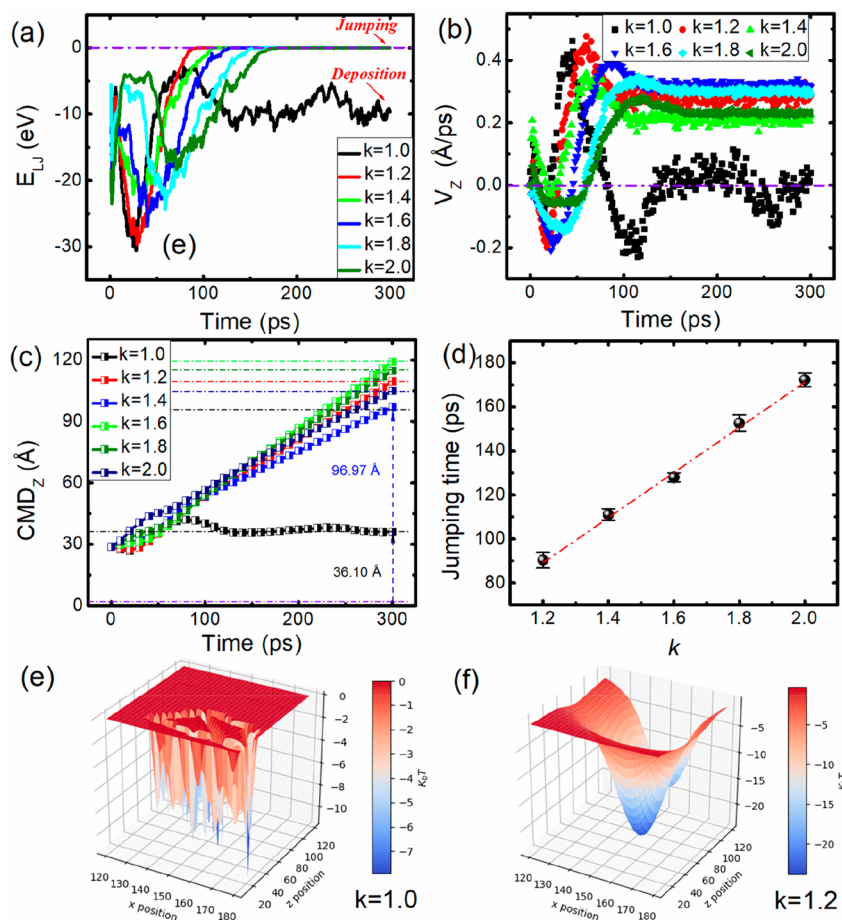


Figure 5. (a) Droplet–substrate interaction energy as a function of time. (b) Movement velocity of the coalesced droplets along the z direction (V_z). (c) Center-of-mass displacement of the Cu droplets along the z direction (CMD_z) with time. (d) Jumping time in different cases when the coalesced droplet starts to detach from the substrate. Free energy landscape based on the x and z position of the selected atoms of the Cu droplets in the case of (e) $k = 1.0$ and (f) $k = 1.2$.

Based on the above studies, it is learned that two J-droplets are able to lift the D-droplet by triple coalescing behaviors. The arrangement for the three droplets in the above-mentioned studies is in a row, where their center-of-mass is in one plane (180°). In many situations, however, the arrangement of the droplets is always random, forming different angles among them.

Therefore, we have carried out another simulation to investigate the effect of arrangement models of the three droplets ($k = 1.2$) on the coalescence-induced jumping behaviors, as shown in Figure 6(a). For the arrangement angles of 66° and 90° , it is unexpected that such a strategy of triple-coalescing-induced jumping behaviors is unsuccessful. Movement velocity curves

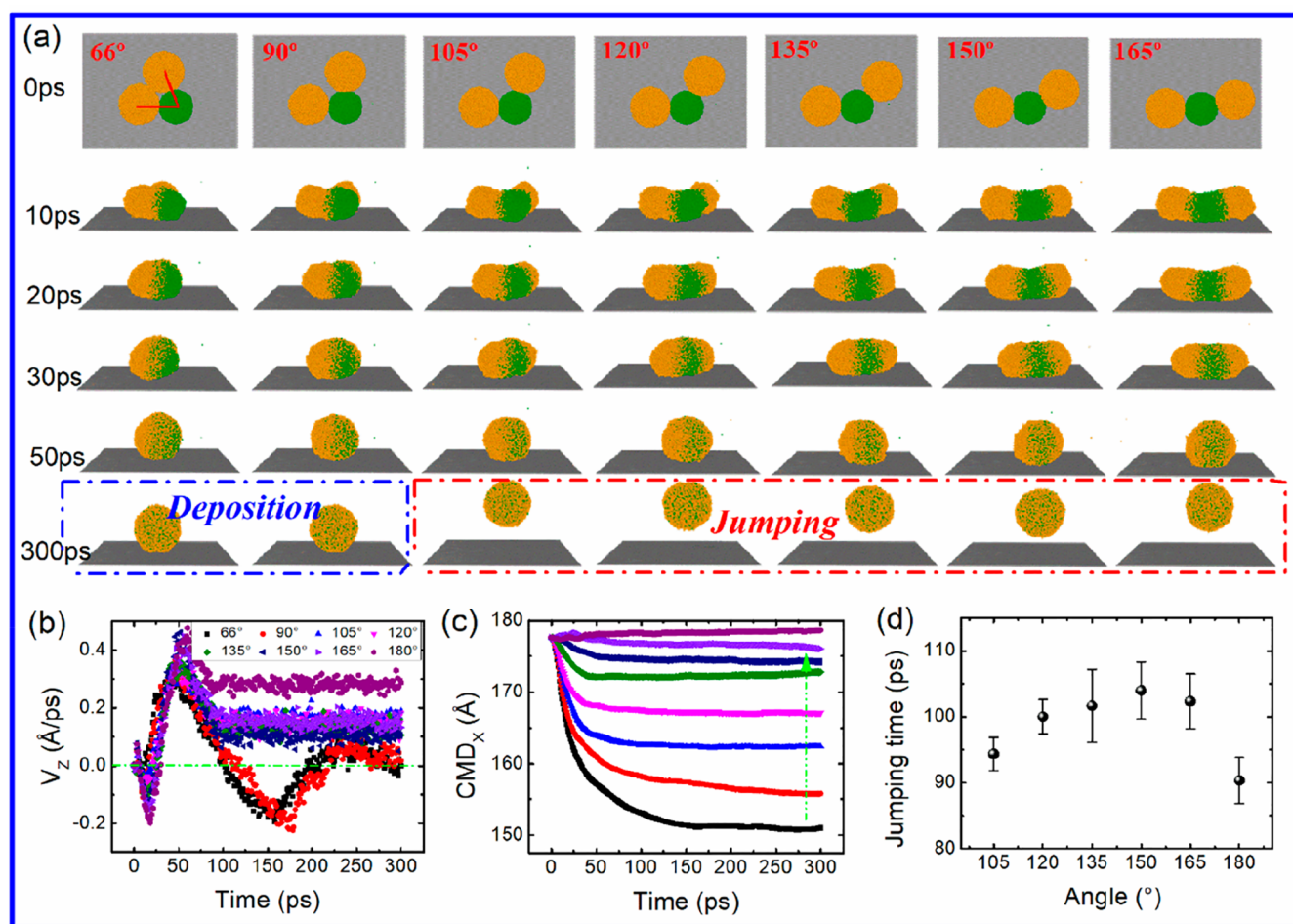


Figure 6. (a) Triple coalescing behaviors of the Cu droplets and two Ag droplets in different arrangement models ($k = 1.2$). (b) Movement velocity of the coalesced droplets along the z direction (V_z). (c) Center-of-mass displacement of the Cu droplets along the x direction (CMD_x) with time. (d) Jumping time of the coalesced droplets in different arrangement models.

(Figure 6(b)) show that the final values of V_z of these two cases nearly becomes zero, indicating that they lose the possibility to detach from the substrates. We also note the V_z for these two cases continues to decrease to values below zero, suggesting that the coalesced droplets move downward during a period of time, which enlarges the contact area between droplets and substrates and thereby increases their interaction. So the coalesced droplets are still left on the surfaces. From 105° to 180°, the D-droplets are all removed with the help of jumping of coalesced droplets, as displayed in Figure 6(a). We also find that their V_z values could not be lower than zero during the decreasing stage, indicating a positive jumping velocity along the z direction. Although these cases can induce the jumping of D-droplets, the jumping time is significantly affected by the arrangement models. In Figure 6(d), the jumping time of the detachment cases is calculated, which shows no monotonous variation with angles. It is found that when the angle is 180, the jumping time is the shortest, implying that three heterogeneous droplets in a row are beneficial to improving the coalescence-induced jumping behaviors. Besides, the arrangement model also affects the diffusion behaviors of the D-droplets during coalescing. Figure 6(c) shows that with the decrease of the angles, the D-droplets would have a larger movement distance from their initial position, which may have some impact on the jumping dynamics.

Figure 7 shows the free energy landscape during the triple coalescing process with four different arrangement models. Here we mainly use it to discuss the effect of angles on the movement of the Cu droplet during coalescing, so the free energy is calculated before the droplets are jumping for all angles, and the range of z position is only set from 10 to 50 Å. From the free energy landscape, we can find that, at small angles, the energy minima mainly gather in the low x position. With the increase of the angles, energy minima gradually move to the larger x position. This indicates the favorable x position of the droplet at different angles, which explains the movement of the droplets along the x directions shown in Figure 6(c) and illustrates the effects of arrangement models on the primary coalescing dynamics that greatly affect the final jumping behaviors.

CONCLUSION

Molecular dynamics simulations have been carried out to study coalescence-induced jumping of heterogeneous droplets of the Ag droplets (J-droplet) and Cu droplets (D-droplet). We first use different sizes of the J-droplets with a radius ranging from 25 to 50 Å to lift a D-droplet with a radius of 25 Å, which fails to remove the D-droplet for all these sizes, but it would work when the lifting ability of J-droplet is enhanced by decreasing the liquid–solid interaction and increasing its size. However, this method for removing the D-droplet is inefficient. Therefore, another strategy that two J-droplets are used to produce triple-

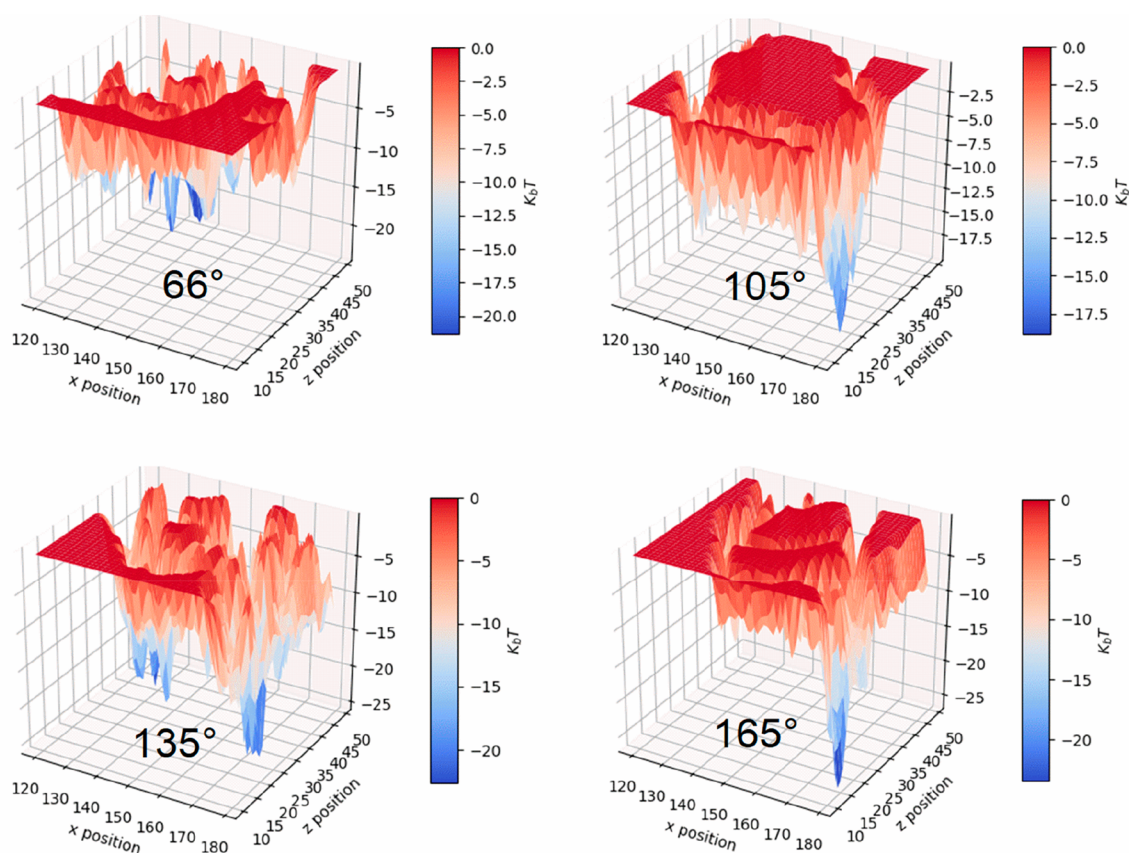


Figure 7. Free energy landscape during the triple coalescing process.

coalescence dynamics for lifting the D-droplets is provided. Here, two J-droplets are on the two sides and the D-droplet is in the center. The arrangement of these three droplets that are in a row (arrangement angle is 180°) is first considered, and simulation results have shown that when $k > 1.2$, the D-droplets could be successfully removed due to such triple-coalescence-induced effect. It is found that with the increase of the size of J-droplet, the jumping time becomes longer. Furthermore, other arrangement models of these three droplets in the case of $k = 1.2$ that produces arrangement angles of 66° , 90° , 105° , 120° , 135° , 150° , 165° are also investigated. The results indicate that closely arranged models (i.e., 66° , 90°) are bad for the jumping of the coalesced droplets, which should be avoided. On the other hand, the jumping time shows no monotonous change with angles and has a smaller value at an angle of 180° , indicating such an arrangement model is favorable for the removal of D-droplets. This work offers a novel view for removing the deposited droplets based on the coalescence-induced jumping dynamics, which may be used for a wide range of applications, especially in the area of self-cleaning.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.2c05570>.

Simulation models for the coalescence dynamics of two heterogeneous droplets (Figure S1), coalescing behaviors of two identical droplets (Figure S2), and the reason for using droplets of silver and copper instead of water-related droplets for the study (PDF)

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Notes

The author declares no competing financial interest.

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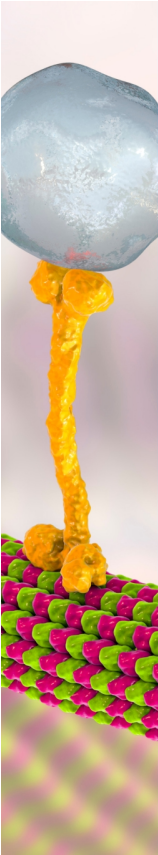
This work was completed using resources provided by The Hong Kong University of Science and Technology's research computing cluster (HPC2).

■ REFERENCES

- (1) Vahabi, H.; Wang, W.; Mabry, J. M.; Kota, A. K. Coalescence-induced jumping of droplets on superomniphobic surfaces with macrotexture. *Sci. Adv.* **2018**, 4 (11), eaau3488.
- (2) Moradi, M.; Rahimian, M. H.; Chini, S. F. Coalescence-induced droplet detachment on low-adhesion surfaces: A three-phase system study. *Phys. Rev. E* **2019**, 99 (6), 063102.
- (3) Liu, C.; Zhao, M.; Zheng, Y.; Cheng, L.; Zhang, J.; Tee, C. A. T. Coalescence-Induced Droplet Jumping. *Langmuir* **2021**, 37 (3), 983–1000.
- (4) Wisdom, K. M.; Watson, J. A.; Qu, X.; Liu, F.; Watson, G. S.; Chen, C.-H. Self-cleaning of superhydrophobic surfaces by self-propelled jumping condensate. *Proc. Natl. Acad. Sci. U. S. A.* **2013**, 110 (20), 7992–7997.

- (5) Li, T.; Li, M.; Li, H. Impact-Induced Removal of the Deposited Droplet: Implications for Self-Cleaning Properties. *J. Phys. Chem. Lett.* **2020**, *11* (15), 6396–6403.
- (6) Yu, Z.; Zhang, K.; Zhao, J.; Chen, S.; Lin, C.; Liu, Y. Coalescence-induced jumping of droplets on superhydrophobic substrates with a beam structure. *Appl. Surf. Sci.* **2022**, *582*, 152284.
- (7) Boreyko, J. B.; Collier, C. P. Delayed frost growth on jumping-drop superhydrophobic surfaces. *ACS Nano* **2013**, *7* (2), 1618–1627.
- (8) Lv, J.; Song, Y.; Jiang, L.; Wang, J. Bio-inspired strategies for anti-icing. *ACS Nano* **2014**, *8* (4), 3152–3169.
- (9) Cha, H.; Xu, C.; Sotelo, J.; Chun, J. M.; Yokoyama, Y.; Enright, R.; Miljkovic, N. Coalescence-induced nanodroplet jumping. *Phys. Rev. Fluids* **2016**, *1* (6), 064102.
- (10) Chen, H.; Ran, T.; Gan, Y.; Zhou, J.; Zhang, Y.; Zhang, L.; Zhang, D.; Jiang, L. Ultrafast water harvesting and transport in hierarchical microchannels. *Nat. Mater.* **2018**, *17* (10), 935–942.
- (11) Tang, Y.; Yang, X.; Li, Y.; Lu, Y.; Zhu, D. Robust micro-nanostructured superhydrophobic surfaces for long-term dropwise condensation. *Nano Lett.* **2021**, *21* (22), 9824–9833.
- (12) Wen, R.; Xu, S.; Zhao, D.; Yang, L.; Ma, X.; Liu, W.; Lee, Y.-C.; Yang, R. Sustaining enhanced condensation on hierarchical mesh-covered surfaces. *Natl. Sci. Rev.* **2018**, *5* (6), 878–887.
- (13) Chen, X.; Wang, P.; Zhang, D.; Ou, J. Rational fabrication of superhydrophobic surfaces with coalescence-induced droplet jumping behavior for atmospheric corrosion protection. *Chem. Eng. J.* **2022**, *428*, 132029.
- (14) Liu, X.; Wang, P.; Zhang, D.; Chen, X. Atmospheric Corrosion Protection Performance and Mechanism of Superhydrophobic Surface Based on Coalescence-Induced Droplet Self-Jumping Behavior. *ACS Appl. Mater. Interfaces* **2021**, *13* (21), 25438–25450.
- (15) Li, T.; Wang, J.; Wang, F.; Zhang, L.; Jiang, Y.; Arandiyani, H.; Li, H. The effect of surface wettability and coalescence dynamics in catalytic performance and catalyst preparation: a review. *ChemCatChem* **2019**, *11* (6), 1576–1586.
- (16) Li, T.; Zhang, L.; Li, H. Controllable Coalescence Dynamics of Nanodroplets on Textured Surfaces Decorated with Well-Designed Wrinkled Nanostructures: A Molecular Dynamics Study. *Langmuir* **2021**, *37* (38), 11414–11421.
- (17) Li, T.; Li, M.; Wang, J.; Li, J.; Duan, Y.; Li, H. Effect of nanopillared surfaces with an arrangement density gradient on droplet coalescence dynamics. *Phys. Chem. Chem. Phys.* **2018**, *20* (38), 24750–24758.
- (18) Li, T.; Xia, Y.; Zhang, L.; Zhang, X.; Fu, C.; Jiang, Y.; Li, H. Effects of stripy surfaces with intervals on the coalescence dynamics of nano-droplets: Insights from molecular dynamics simulations. *Appl. Surf. Sci.* **2019**, *481*, 951–959.
- (19) Li, T.; Zhang, L.; Zhang, X.; Li, H. Effect of curved surfaces on the impacting nano-droplets and their shape control: A molecular dynamics simulation study. *Appl. Surf. Sci.* **2018**, *454*, 192–200.
- (20) Gao, S.; Liao, Q.; Liu, W.; Liu, Z. Coalescence-induced jumping of nanodroplets on textured surfaces. *J. Phys. Chem. Lett.* **2018**, *9* (1), 13–18.
- (21) He, X.; Zhao, L.; Cheng, J. Coalescence-induced swift jumping of nanodroplets on curved surfaces. *Langmuir* **2019**, *35* (30), 9979–9987.
- (22) Lu, D.; Zhao, M.; Zhang, H.; Yang, Y.; Zheng, Y. Self-enhancement of coalescence-induced droplet jumping on superhydrophobic surfaces with an asymmetric v-groove. *Langmuir* **2020**, *36* (19), 5444–5453.
- (23) Han, T.; Kwak, H. J.; Kim, J. H.; Kwon, J.-T.; Kim, M. H. Nanograsped zigzag structures to promote coalescence-induced droplet jumping. *Langmuir* **2019**, *35* (27), 9093–9099.
- (24) Chen, X.; Weibel, J. A.; Garimella, S. V. Characterization of coalescence-induced droplet jumping height on hierarchical superhydrophobic surfaces. *ACS omega* **2017**, *2* (6), 2883–2890.
- (25) Chen, X.; Patel, R. S.; Weibel, J. A.; Garimella, S. V. Coalescence-induced jumping of multiple condensate droplets on hierarchical superhydrophobic surfaces. *Sci. Rep.* **2016**, *6* (1), 18649.
- (26) Cheng, Y.; Du, B.; Wang, K.; Chen, Y.; Lan, Z.; Wang, Z.; Ma, X. Macrotextures-induced jumping relay of condensate droplets. *Appl. Phys. Lett.* **2019**, *114* (9), 093704.
- (27) Li, T.; Wu, Y. Impact Dynamics of Nanodroplets on V-Shaped Substrates: Asymmetrical Behavior and Fast-Rebound Dynamics. *Langmuir* **2021**, *37* (44), 13170–13178.
- (28) Wang, F.-C.; Yang, F.; Zhao, Y.-P. Size effect on the coalescence-induced self-propelled droplet. *Appl. Phys. Lett.* **2011**, *98* (5), 053112.
- (29) Wang, K.; Li, R.; Liang, Q.; Jiang, R.; Zheng, Y.; Lan, Z.; Ma, X. Critical size ratio for coalescence-induced droplet jumping on superhydrophobic surfaces. *Appl. Phys. Lett.* **2017**, *111* (6), 061603.
- (30) Wasserfall, J.; Figueiredo, P.; Kneer, R.; Rohlf, W.; Pischke, P. Coalescence-induced droplet jumping on superhydrophobic surfaces: Effects of droplet mismatch. *Phys. Rev. Fluids* **2017**, *2* (12), 123601.
- (31) Xie, F.-F.; Lu, G.; Wang, X.-D.; Wang, B.-B. Coalescence-induced jumping of two unequal-sized nanodroplets. *Langmuir* **2018**, *34* (8), 2734–2740.
- (32) Li, T.; Li, J.; Wang, L.; Duan, Y.; Li, H. Coalescence of Immiscible Liquid Metal Drop on Graphene. *Sci. Rep.* **2016**, *6*, 34074.
- (33) Plimpton, S. Fast parallel algorithms for short-range molecular dynamics. *J. Comput. Phys.* **1995**, *117* (1), 1–19.
- (34) Shinoda, W.; DeVane, R.; Klein, M. L. Multi-property fitting and parameterization of a coarse grained model for aqueous surfactants. *Mol. Simul.* **2007**, *33* (1–2), 27–36.
- (35) Shinoda, W.; DeVane, R.; Klein, M. L. Coarse-grained molecular modeling of non-ionic surfactant self-assembly. *Soft Matter* **2008**, *4* (12), 2454–2462.
- (36) Martys, N. S.; Mountain, R. D. Velocity Verlet algorithm for dissipative-particle-dynamics-based models of suspensions. *Phys. Rev. E* **1999**, *59* (3), 3733.
- (37) Wang, S.; Fan, Z.; Cui, Y.; Zhang, S.; Yang, B.; Chen, H. Fracture behaviors of brittle and ductile 2D carbon structures under uniaxial tensile stress. *Carbon* **2017**, *111*, 486–492.
- (38) Lee, G.-D.; Yoon, E.; He, K.; Robertson, A. W.; Warner, J. H. Detailed formation processes of stable dislocations in graphene. *Nanoscale* **2014**, *6* (24), 14836–14844.
- (39) Qi, L.; Mao, Y.; Li, J. Slip corona surrounding bilayer graphene nanopore. *Nanoscale* **2012**, *4* (19), 5989–5997.
- (40) Wu, H. H.; Trinkle, D. R. Cu/Ag EAM potential optimized for heteroepitaxial diffusion from ab initio data. *Comput. Mater. Sci.* **2009**, *47* (2), 577–583.
- (41) Li, T.; Wang, Z.; Duan, Y.; Li, J.; Li, H. Molecular dynamics study on the formation of self-organized core/shell structures in the Pb alloy at the nanoscale. *RSC Adv.* **2017**, *7* (84), 53509–53515.
- (42) Li, T.; Zhang, L.; Duan, Y.; Li, J.; Wang, Z.; Wang, J.; Li, H. Bouncing dynamics of liquid drops impact on ridge structure: An effective approach to reduce the contact time. *Phys. Chem. Chem. Phys.* **2018**, *20* (24), 16493–16500.
- (43) Fuentes-Cabrera, M.; Rhodes, B. H.; Baskes, M. I.; Terrones, H.; Fowlkes, J. D.; Simpson, M. L.; Rack, P. D. Controlling the velocity of jumping nanodroplets via their initial shape and temperature. *ACS Nano* **2011**, *5* (9), 7130–7136.
- (44) Sidky, H.; Colón, Y. J.; Helfferich, J.; Sikora, B. J.; Bezik, C.; Chu, W.; Giberti, F.; Guo, A. Z.; Jiang, X.; Lequeieu, J.; et al. SSAGES: software suite for advanced general ensemble simulations. *J. Chem. Phys.* **2018**, *148* (4), 044104.
- (45) Laio, A.; Parrinello, M. Escaping free-energy minima. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99* (20), 12562–12566.
- (46) Zeng, D.; Li, Y.; Huan, D.; Liu, H.; Luo, H.; Cui, Y.; Zhu, C.; Wang, J. Robust epoxy-modified superhydrophobic coating for aircraft anti-icing systems. *Colloids Surf., A* **2021**, *628*, 127377.
- (47) Lin, Y.; Chen, H.; Wang, G.; Liu, A. Recent progress in preparation and anti-icing applications of superhydrophobic coatings. *Coatings* **2018**, *8* (6), 208.
- (48) Sun, Z.; Liao, T.; Liu, K.; Jiang, L.; Kim, J. H.; Dou, S. X. Fly-eye inspired superhydrophobic anti-fogging inorganic nanostructures. *Small* **2014**, *10* (15), 3001–3006.
- (49) Lin, S.; Yang, Y.; Ni, J.; Tsurimaki, Y.; Liu, X.; Lu, B.; Tu, Y.; Zhou, J.; Zhao, X.; Chen, G. Stretchable Anti-Fogging Tapes for

Diverse Transparent Materials. *Adv. Funct. Mater.* **2021**, 31 (36), 2103551.



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